

Assignment 1

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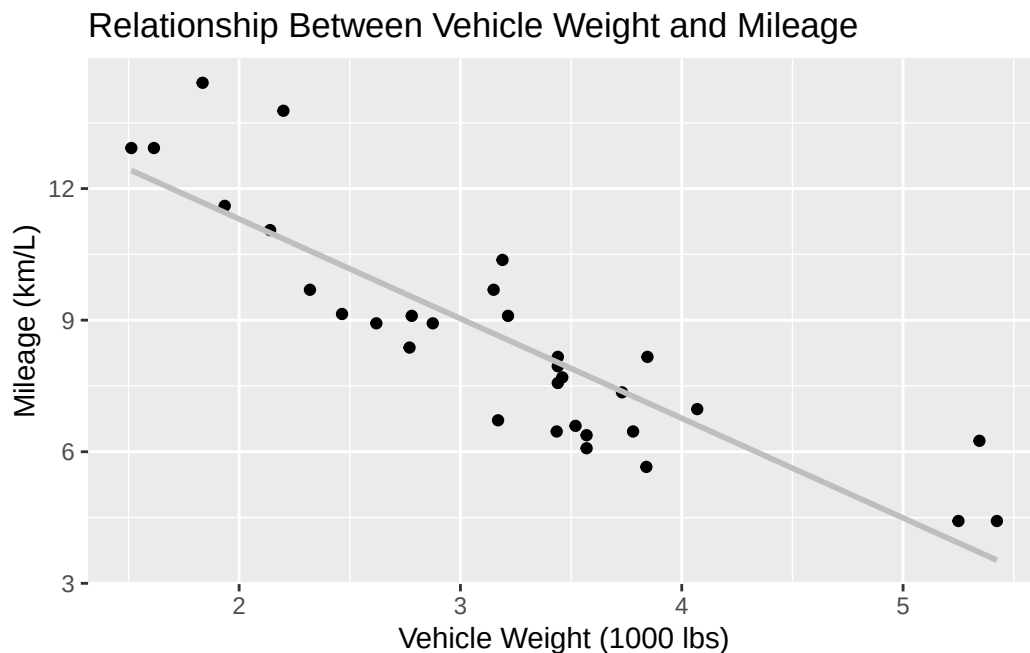
Research Question

Does the mileage of a car increase as its weight increases?

Answer

No, the mileage of a car decreases as its weight increases. To address this question, I used following procedures:

1. Subset the *mtcars* dataset into a new one with only two columns (mpg and wt).
2. Convert the miles per gallon into kilometres per litre. (1mile=1.60934 kilometres, 1 US gallon=3.78541 litres), then $\text{km/L} = \text{m/g} * 1.60934 / 3.78541$
3. Generate a scatter plot using `geom_point` in `ggplot2` and add a regression line using `geom_smooth`.



4. The scatter plot suggests a negative linear relationship between vehicle weight and mileage.